

Seongmoon Jeong

PHD CANDIDATE

IRIS Lab., Sungkyunkwan University, Suwon, Republic of Korea

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I am a Ph.D. candidate in the Department of Artificial Intelligence at Sungkyunkwan University, advised by **Prof. Jong Hwan Ko**. My research focuses on **image compression for machine perception**, spanning low-level optimization and high-level tasks such as classification, object detection, and segmentation. I develop **learned, task-aware representations** that jointly optimize compression efficiency and downstream vision performance. More recently, I have investigated high-level vision directly in the **RAW domain**, before the image signal processing (ISP) pipeline, to preserve information relevant to machine perception. I have authored or co-authored **eight peer-reviewed publications**, including papers at CVPR, AVSS, ICCAD, and in IEEE TCAS-I, and contributed **four ISO/IEC MPEG VCM proposals**. My long-term goal is to enable **communication-efficient intelligent systems** for autonomous systems and edge-based machine vision. **Available for full-time roles starting March 2027.**

Education

Ph.D. in Artificial Intelligence (Expected February 2027)

SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2020 - Present

- Received an Outstanding Scholarship awarded to promising students in the Department of Artificial Intelligence.

B.S. in Electronic and Electrical Engineering

SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2013 - Feb. 2020

- GPA: 3.84/4.5 (Top 6.6%)

Research Experience

Graduate Researcher

IRIS LAB, SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2020 - Present

- Machine-Oriented Image Compression:** Codec-transferable and test-time adaptation across downstream vision tasks [C4, C5]
- Task-Aware Preprocessing and JPEG Coding:** Adaptive downscaling, feature modulation, and region-based JPEG optimization [C1, C3, J1]
- RAW-Domain Machine Vision:** Foundation-model-guided RGB-to-RAW generation for object detection [C6]

Project Experience

Video Coding for Machines

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU)

Media Research Division, ETRI

2023 - 2024

- Led research on codec-transferable adaptation for machine-oriented image compression [C5].

Adaptive and Efficient Preprocessing and Coding for Machine Vision Tasks and Input Images

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU)

Media Research Division, ETRI

2022

- Led research on adaptive image downscaling for rate-accuracy-latency optimization [C3].

Multipurpose Visual Information Compression for Human and Machine Vision

PARTICIPATING RESEARCHER (BASIC RESEARCH LABORATORY, IRIS LAB., SKKU)

Ministry of Science and ICT (MSIT),
Korea

2025 - 2028

- Contributing to unified visual compression methods supporting both human viewing and machine perception.

Selected Publications

C6 Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection

Seongmoon Jeong, Yulhwa Kim, Jong Hwan Ko

IEEE AVSS (Accepted)

Sep. 2026

C5 UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression

Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Hyon-Gon Choo, Jong Hwan Ko

IEEE AVSS (Accepted)

Sep. 2026

C4 Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability

Un Ki Park, Seongmoon Jeong, Youngchan Jang, Gyeongmoon Park, Jong Hwan Ko

IEEE CVPR

Mar. 2025

C3 Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

Hangyul Choi, Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Jong Hwan Ko

IEEE AICAS

Feb. 2024

Standardization Activities

[VCM] Neural network based post-filter for VCM

ISO/IEC JTC1 / SC29 / WG4 M71305

Geneva

Jan. 2025

[VCM] Neural network based pre-filter for VCM

ISO/IEC JTC1 / SC29 / WG4 M71304

Geneva

Jan. 2025

[VCM] Learned joint filter network for VCM

ISO/IEC JTC1 / SC29 / WG4 M69990

Kemer

Nov. 2024

[VCM] Learned pre-filter network for VCM

ISO/IEC JTC1 / SC29 / WG4 M69085

Sapporo

Jan. 2024

Honors & Awards

2024	3rd Place , DAC System Design Contest - GPU Track	San Francisco, USA
2020	2nd Place , Artificial Intelligence Grand Challenge, Ministry of Science and ICT	Korea
2019	Honorable Mention , Big Data Center Open Innovation Challenge	Seongnam, Korea
2019	Outstanding Work Award , Capstone Design Project, SKKU	Suwon, Korea
2018	Dean's List Award , Fall semester, SKKU	Suwon, Korea
2017	Dean's List Award , Fall semester, SKKU	Suwon, Korea

Full Publications

INTERNATIONAL CONFERENCE

C6 Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection

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Hangyul Choi, Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Jong Hwan Ko

IEEE AICAS

Feb. 2024

C2 Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays

Johnny Rhe, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, Jong Hwan Ko

IEEE ICCAD

Jul. 2023

C1 Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation

Seongmoon Jeong, Kang Eun Jeon, Jong Hwan Ko

IEEE ICME Workshop

Apr. 2023

INTERNATIONAL JOURNAL

J2 KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping

Johnny Rhe, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, Jong Hwan Ko

IEEE TCAS-I

Feb. 2024

J1 An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression

Seonghye Jeong, Seongmoon Jeong, Simon S. Woo, Jong Hwan Ko

Pattern Recognition Letters

Nov. 2022

Student Mentoring

Hangyul Choi

M.S. STUDENT AT SUNGKYUNKWAN UNIVERSITY, NOW AT MX BUSINESS, SAMSUNG ELECTRONICS

2023 - 2024

- Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression [C3]

Chanung Park

GRADUATE STUDENT AT SUNGKYUNKWAN UNIVERSITY

2024 - Present

- Research collaboration on Video Coding for Machines (VCM) and the BRL project on multipurpose visual information compression for human and machine vision.

Skills

Programming Languages	Python, Shell scripting, C/C++
ML Frameworks	PyTorch, TensorFlow, JAX, TensorRT
Development Frameworks	Docker, Conda, uv

Reference

Jong Hwan Ko

ASSOCIATE PROFESSOR

Ph.D. advisor

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